

Kourosh Sajjadi

+98 912 068 3805 • kouroshsajjadi@yahoo.com • ksajjadi.github.io
linkedin.com/in/kourosh-sajjadi • github.com/Ksajjadi

Education

University of Tehran

B.Sc. in Computer Engineering (Software Engineering Concentration),
School of Electrical and Computer Engineering, College of Engineering

Tehran, Iran

Sep. 2021 – Expected Sep. 2026

- GPA: **18.91/20 (3.93/4.00)**; ranked among the **top 5%** of B.Sc. students in Computer Engineering.

Selected coursework

- Neural Networks and Deep Learning (20/20)
- Fundamentals of Computer Vision (20/20)
- Compiler Design and Programming Languages (20/20)
- Artificial Intelligence (20/20)
- Fundamentals of Data Science (20/20)
- Data Structures and Algorithms (20/20)

Research interests

- AI Evaluation & Reliability
- Computer Vision & Multimodal Learning
- Reliable LLM Systems & Agents
- Robustness under Distribution Shift
- AI for Software Engineering
- AI in Education & Intelligent Tutoring

Research experience

Undergraduate Researcher - B.Sc. Thesis

TaD-REV: Task- and Domain-Grounded, Evidence-Validated Stateful Review of Code Changes

Oct. 2025 – Present

University of Tehran

- Built a commit-level LLM code-review pipeline grounded in task specifications, rubrics, domain documents, and repository baselines; it preserves first-pass and evidence-validated outputs as structured artifacts for each reviewed commit.
- Designed an F0/I0/I1 study over authentic University of Tehran histories with stable finding IDs, a validated cross-commit ledger, full telemetry, and a 75-finding double-annotated sample. Supervised by Prof. Ramtin Khosravi; manuscript targeted for Sep./Oct. 2026.

Undergraduate Researcher

HARBOR: Post-Earthquake Building Damage Assessment from Pre-Event Optical and Post-Event SAR Imagery

Oct. 2025 – Present

University of Tehran

- Benchmarked U-Net and ChangeMamba on 373 earthquake test tiles; building localization reached 91.56% F1 while damage classification fell to 32.77%, with 74.4% of damaged and 54.8% of destroyed buildings predicted as intact.
- Designed a three-stage architecture with an affected-building gate, reliability-gated optical-SAR alignment, and object-region pooling. Drive the study's architecture design and experiments in collaboration with Dr. Ahmad Toosi, under Prof. Farhad Samadzadegan; first-author manuscript targeted for Sep./Oct. 2026.

Research Assistant

LLM-Simulated Persian Students for Evaluating AI Tutoring and Personalization

Feb. 2026 – Present

University of Tehran

- Designed LLM-based Persian student personas with bounded knowledge, stable misconceptions, confidence, engagement, and help-seeking behavior to evaluate an AI tutor before enough real dialogue is available.
- Building a black-box harness for 5–10-turn LMS sessions that records transcripts and metadata for persona consistency, misconception stability, and tutor-strategy analysis. The output will be a controlled dialogue dataset for evaluation and future personalization. Supervised by Prof. Majid Nili Ahmadabadi; completion date TBD.

Languages

English: Full professional proficiency; IELTS Academic 8.0
(CEFR C1; Listening 9.0, Reading 8.5, Speaking 8.0, Writing 7.0), Oct. 2025

Persian: Native

Honors and awards

- Faculty of Engineering Award:** University of Tehran; recipient in 2021–2022 and nominee in the following cycle.
- National entrance examination:** Ranked top 0.5% in the Iranian Nationwide University Entrance Examination.
- Benefactors Foundation Scholarship:** 8th University of Tehran scholarship for talented students, 2022–2023.
- Academic standing:** Ranked among the top 5% of B.Sc. students in Computer Engineering.

Teaching experience

Teaching Assistant (programming assignments, tutorials, and homework/exam grading) <i>School of Electrical and Computer Engineering, University of Tehran</i>	2023 – 2026 <i>Tehran, Iran</i>
Artificial Intelligence	Spring 2026, Fall 2025, Spring 2025
Data Structures and Algorithms	Fall 2024, Spring 2024, Fall 2023
Principles of Database Design	Fall 2025, Spring 2025
Compiler Design and Programming Languages	Fall 2025, Fall 2024
Computer Networks	Spring 2025, Fall 2025

Professional experience

Part-Time Full-Stack Developer and AI Engineer <i>RastAI, University of Tehran Science and Technology Park</i>	Nov. 2025 – Present <i>Tehran, Iran</i>
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- Develop Kotlin and Spring backend services, Next.js and React interfaces, and Python AI components for an LLM-powered personalized learning platform.
- Design agentic workflows with the OpenAI Agents SDK, LangChain, and Langfuse, covering API integration, tracing, evaluation, and observability; work across the product team to turn learning requirements into implementable system designs.

Computer Vision Intern <i>Private Company, University of Tehran</i>	Summer 2024 <i>Tehran, Iran</i>
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- Built Python computer-vision applications covering image processing, OCR, classification, object detection, pose estimation, and segmentation using OpenCV, MediaPipe, EasyOCR, YOLOv8, scikit-learn, Segment Anything, Streamlit, Stable Diffusion, and DeepFace.

Selected projects

Deep Learning Paper Reproductions | *Python, PyTorch, TensorFlow* | Repository

- LoRA fine-tuned the 8B-parameter LLaDA masked-diffusion model for Text-to-SQL and implemented its iterative sampling loop from scratch; reached 17.3% normalized exact match on 300 held-out queries.  Code
- Reproduced INC-VGGN for plant-disease classification at 92.24% accuracy versus 92.00% reported; reproduced M-Net for brain-MRI tissue segmentation and cleared all three Dice targets; implemented a VampPrior VAE study of posterior collapse.  M-Net |  VAE

Distributed Systems Course Projects | *Go, RPC, MapReduce, Raft* | Repository

- Implemented RPC services, MapReduce, Raft consensus, and sharded key-value stores in Go with tests and documentation based on MIT 6.5840/6.824-style labs.

Bird's-Eye-View Stitching System | *C++, OpenCV, Qt, Raspberry Pi* | Repository

- Built a real-time surround-view simulator with interactive calibration, an image-stitching path measured at approximately 23 ms per frame, and support for embedded deployment.

Spotify Song Recommendation System | *Python, SQLite, MLflow, Power BI* | Repository

- Built a content-based KNN recommendation pipeline spanning exploratory analysis, feature engineering, reproducible data processing, experiment tracking, persisted model artifacts, and dashboarding.

Software Engineering Matching Engine | *Java, Spring, Artemis JMS, JUnit* | Repository

- Extended a trading matching engine across several iterations with asynchronous messaging, new order types, automated tests, and design-pattern-based restructuring.

Technical skills

Programming: Python, C, C++, Java, Kotlin, Go, JavaScript, TypeScript, SQL, MATLAB, Verilog, C#

AI and machine learning: PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy, OpenCV, YOLOv8, Hugging Face, LangChain, Haystack, OpenAI Agents SDK

Web and backend: Spring Boot, Next.js, React, FastAPI, REST APIs, PostgreSQL, MongoDB, SQLite, Redis

Data and infrastructure: Docker, Git, Linux, Kafka, MLflow, Langfuse, Jupyter, LaTeX, JUnit, pytest